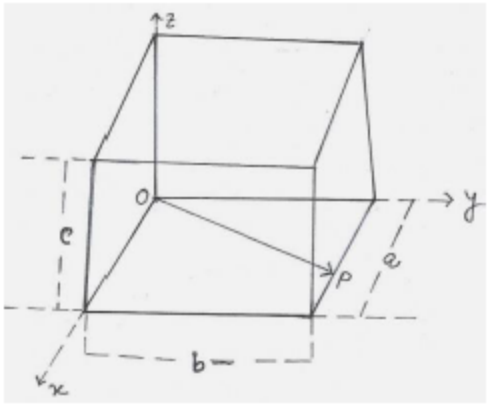
Solid State Physics — Week 2 Assignment Questions & Answers

Question 11

Determine the indices for the direction $(\overrightarrow{OP})$ shown in the accompanying figure.



(a) [120]

(b) [201]

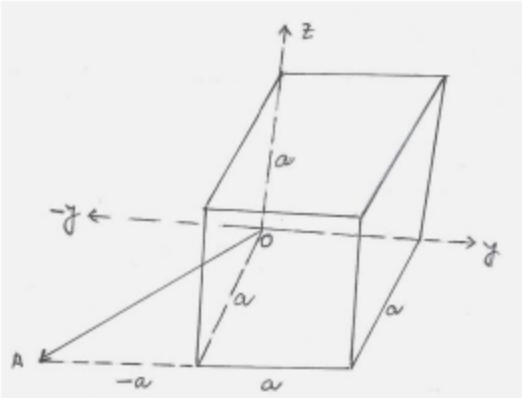
(c) [102]

(d) [210]

Accepted answer: (a)

Question 12

Determine the indices for the direction $\overrightarrow{OA}$ within a cubic unit cell having lattice parameter a, as shown in figure below.



(a)$[\bar{1}\ 10]$

(b)$[1\bar{1}0]$

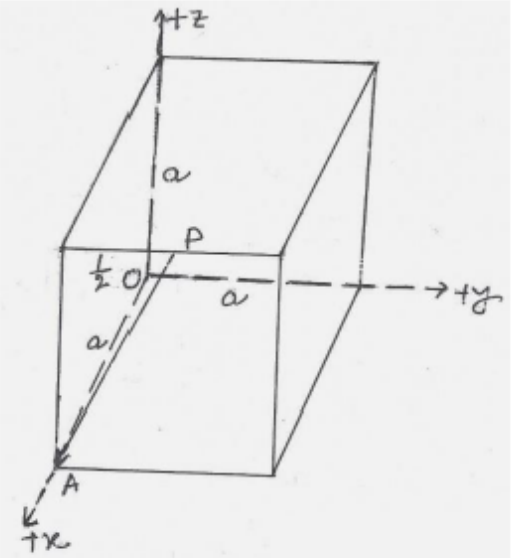
(c)$[01\bar{1}]$

(d) $[10\bar{1}]$

Accepted answer: (b)

Question 13

Determine the indices for the direction $\overrightarrow{PA}$ in a cubic unit cell having edge length a. Accompanying figure as shown in below.



(a) $[0\bar{1}\bar{2}]$

(b) $[\bar{1}0\bar{2}]$

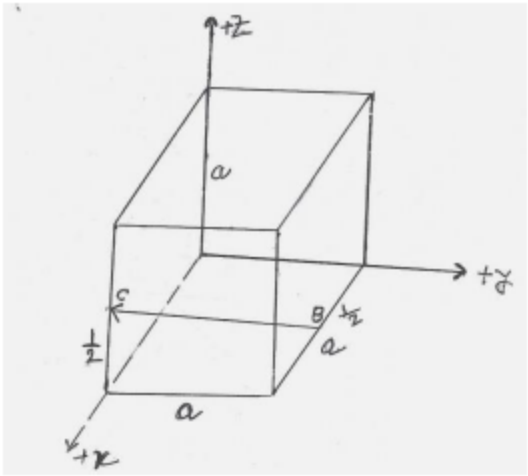
(c) $[\bar{2}\bar{1}0]$

(d) $[\bar{1}\bar{2}0]$

Accepted answer: (a)

Question 14

Determine the indices for direction $\overrightarrow{BC}$ shown in the following cubic unit cell and edge length a. The accompanying figure is given below.



(a) $[1\bar{2}1]$

(b) $[\bar{1}\bar{2}\bar{1}]$

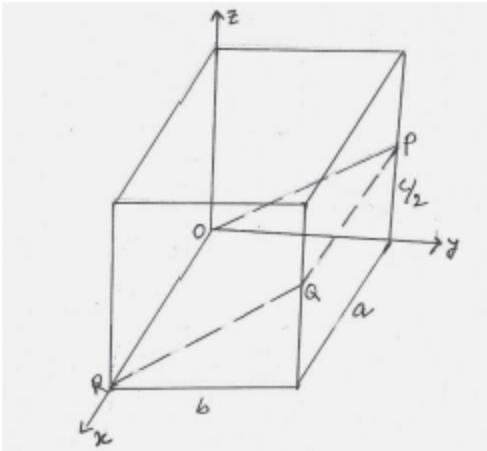
(c) $[\bar{2}11]$

(d) $[1\bar{2}\bar{1}]$

Accepted answer: (a)

Question 15

Determine the Miller indices for the plane (OPQR) as shown in the figure below.



(a) $(0\bar{1}2)$

(b) $(01\bar{2})$

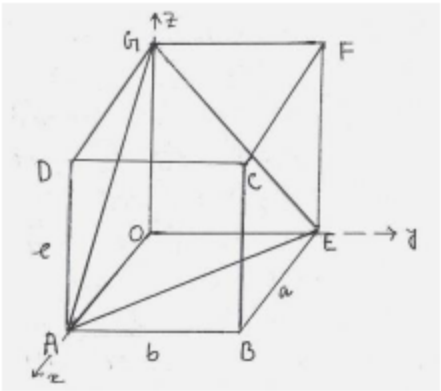
(c) $(\bar{1}1\bar{2})$

(d) $(00\bar{1})$

Accepted answer: (a)

Question 16

Find out the Miller indices of the plane AEG in an orthorhombic lattice as shown in figure below.



(a) (121)

(b) (111)

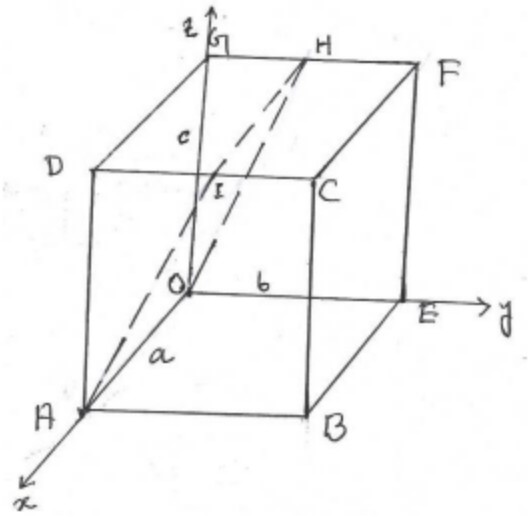
(c) (222)

(d) (112)

Accepted answer: (b)

Question 17

Find out the Miller indices of the plane (OAIH) in an orthorhombic lattice as shown in figure below.



(a) $(0\bar{2}1)$

(b) $(02\bar{1})$

(c) $(0\bar{2}\bar{1})$

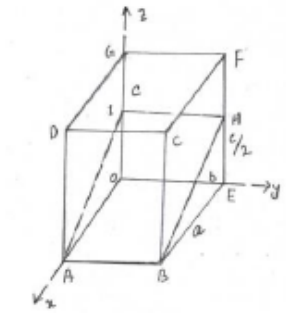
(d) (121)

Accepted answer: (b)

Question 18

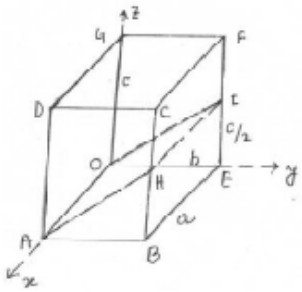
Choose the correct plane corresponding to the Miller indices (102) in an orthorhombic lattice.

(a)



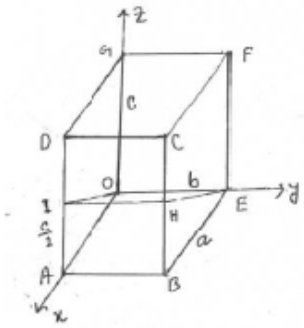
(ABHI) plane

(b)



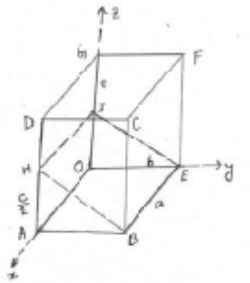
(AHIO) Plane

(c)



(OEHI) Plane

(d)

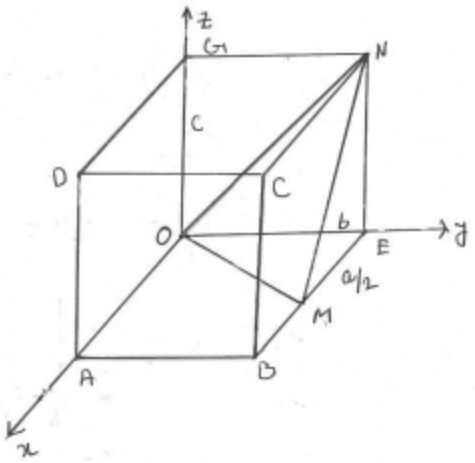


(BEIH) Plane

Accepted answer: (a)

Question 19

Find out the Miller indices of the plane OMN given in the accompanying figure.



(a) $(1\bar{2}1)$

(b) $(2\bar{1}1)$

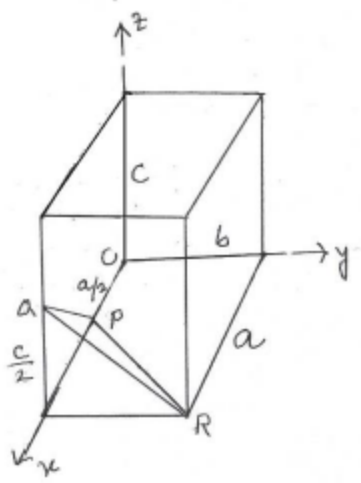
(c) (122)

(d) $(1\bar{2}\bar{1})$

Accepted answer: (b)

Question 20

Find out the Miller indices of the plane (PQR) in the accompanying figure.



(a) $(\bar{2}12)$

(b) $(\bar{2}\bar{1}2)$

(c) $(2\bar{1}2)$

(d) $(21\bar{2})$

Accepted answer: (a)